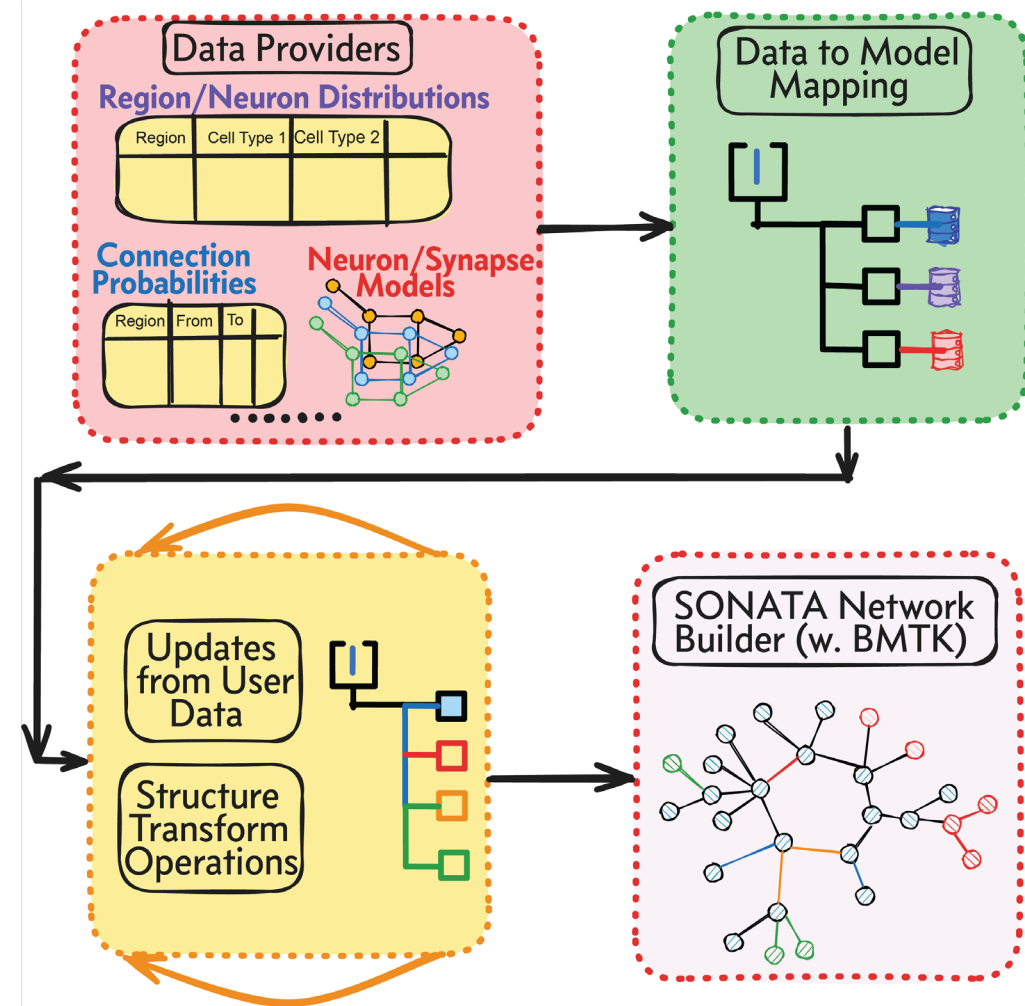
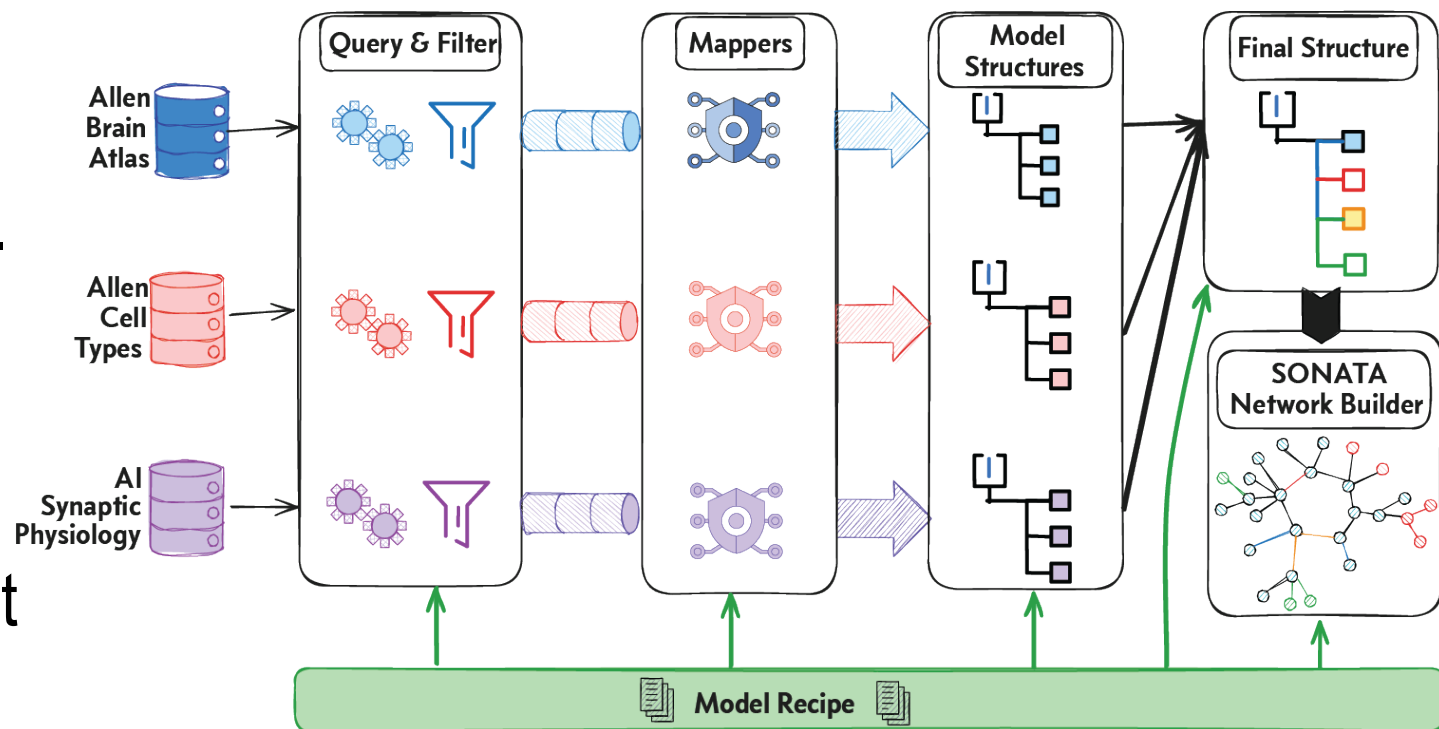


- Large public databases provide extensive and detailed maps of the brain (e.g. Brain Atlas) .
- Using such atlases for the construction of computational models remains a challenge.
- Diverse databases are required for construction of detailed models.
 - Spatial transcriptomics data allows for regional and layer-wise distributions of the different cell types.
 - Synaptic physiology databases provide connection probabilities between the sets of neuron classes.
- Data from diverse sources incorporated to construct detailed models
- *Airavata Cerebrum*
 - Flexible and extendable software that allows queries over different databases and brain atlases.
 - Applies transformations such as maps and filters on the resultant data.
 - Incorporate results within the computational models described using SONATA format

- Extract data from experimental and brain atlas datasets via querying the relevant databases.
- Map acquired data to regions of interest.
- Fill-in any missing details via an optional user-defined custom modifications.
- Realize the final network of neurons in SONATA format from the model recipe.

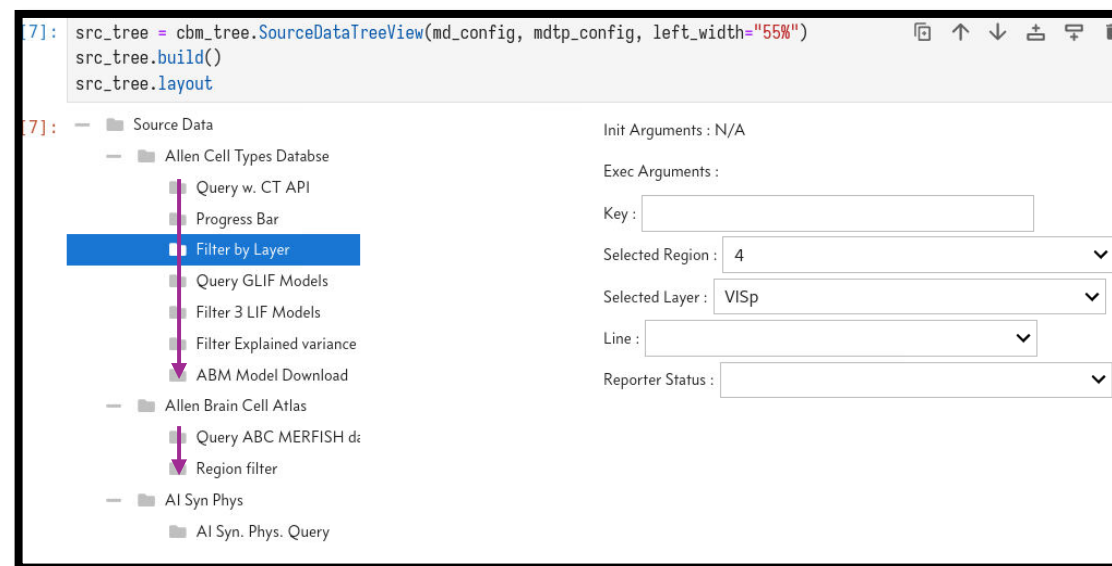


- ❖ **Data Providers:** Links to source database and query parameters
- ❖ **Filter and Transformers:** Filter via selection criteria on the queried data.
- ❖ **Data2Model Mappers:** Mapper methods that map data to model components.
- ❖ **User Modifications:** User Inputs that modify the data-derived model



Show Demo

- Methods for querying from the databases and workflow to process the data based on specific criteria.
- Visualize the data provider configurations in easy-to-understand explorer view inside Jupyter notebook.
- E.g. Layer 4 of Mouse V1
 - Allen Cell Types Database
 - Allen Brain Cell Atlas
 - Allen Inst. Synaptic Physiology Database.



- Custom python functions to map data queried to the different components of the models.
- Ability to integrate different data sources that map to a component in the network model.
- Two different mappings: Region Mapper and Connection Mapper.
- E.g. Layer 4 of Mouse V1
 - **Locations/Regions:**
 - Cell types database $\mapsto$ Neuron models
 - MERFISH atlas data $\mapsto$ Distribution of neurons.
 - **Connections:** AI synaptic physiology data is mapped to the connection probabilities

```
[9]: loc_tree = cbm_tree.D2MLocationsTreeView(md_config, mdtp_config, width="55%")
loc_tree.build()
loc_tree.layout
```

[9]: — locations

- VSp4
 - + property_map
 - + ('4', 'Scnn1a')
 - + ('4', 'Rorb')
 - ('4', 'Vip')
 - Select Cell Types
 - Select Line
 - Filter Cell Fraction
 - Select Region,
 - + ('4', 'Pvalb')
 - + ('4', 'Sst')

Init Arguments: N/A

Exec Arguments:

Key:

Selected Region:

Selected Layer:

Line:

Reporter Status:

```
[10]: conn_tree = cbm_tree.D2MConnectionsTreeView(md_config, mdtp_config)
conn_tree.build()
conn_tree.layout
```

[10]: — Connections

- + (('4', 'Pvalb'), ('4', 'Pvalb'))
- (('4', 'Pvalb'), ('4', 'Sst'))
- AI Syn Phys
- Filter by Pair
- + (('4', 'Pvalb'), ('4', 'Vip'))
- + (('4', 'Sst'), ('4', 'Pvalb'))
- + (('4', 'Sst'), ('4', 'Sst'))
- + (('4', 'Sst'), ('4', 'Vip'))
- + (('4', 'Vip'), ('4', 'Pvalb'))
- + (('4', 'Vip'), ('4', 'Sst'))
- + (('4', 'Vip'), ('4', 'Vip'))

Init Arguments: N/A

Exec Arguments:

Pre-synaptic Type:

Post-synaptic Type:

- Details for network construction that are either not available in the linked databases or over-ride data-derived model.
- Visualize modifications in Jupyter Notebook.
- E.g. Layer 4 of Mouse V1
 - Dimensions of each region in the model.
 - Connection Parameters not available with the Allen Inst. Syn. Phys. database.
 - Details of the networks that are external to V1 – LGN and the background networks.

```
[19]: custom_mods = model_dex.custom_mod_struct()
      cmod_view = cbm_tree.NetworkTreeView(custom_mods, left_width="55%")
      cmod_view.build()
      cmod_view.layout
```

```
[19]:
```

(**'4', 'Vip'**)

Name: (**'4', 'Vip'**)

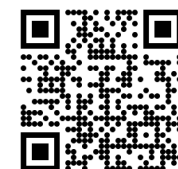
N: 0

Proportion: 0

E/I: i

Property Map: {
 "nsyn_lognorm_shape": 0.537,
 "nsyn_lognorm_scale": 2355.279
 }

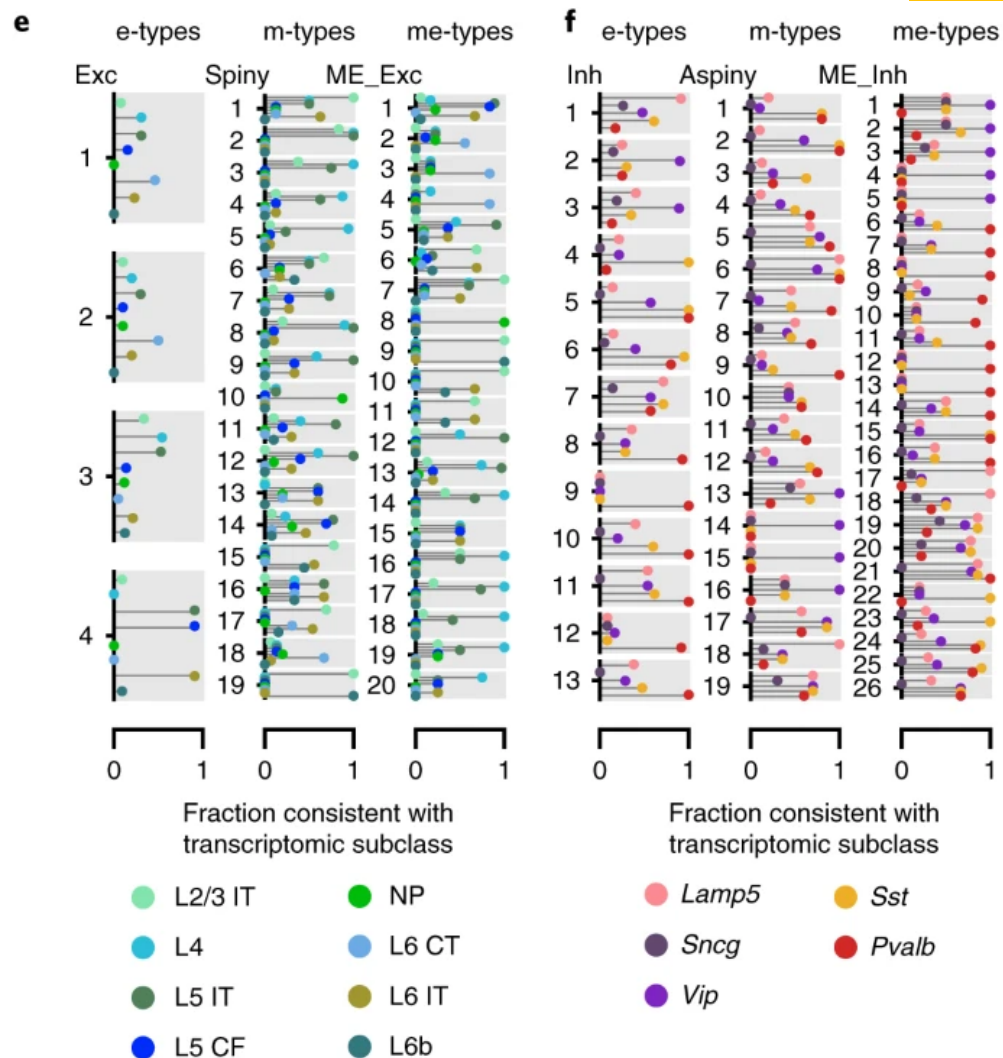
Source: github.com/apache/airavata-cerebrum
 V1L4 model Accessible via cybershuttle.org



Additional Slides

Correspondence between types in Gouwens et. al. (2019)

- Consistency score between E-type, M-type and ME-type vs t-type (transcriptomic type)
- No one-to-one correspondence
- Higher consistency between Inhibitory and ME-type



Classification of Cell Types (Gouwens et. al. 2019)

- Based on morphological and electrophysiological types
- Classified based on
 - E-type
 - 4 Exc and 13 Inh types
 - M-type
 - 19 Exc and 19 Inh type
 - ME-type
 - 20 Exc and 26 Inh types

Resource | [Published: 17 June 2019](#)

Classification of electrophysiological and morphological neuron types in the mouse visual cortex

[Nathan W. Gouwens](#), [Staci A. Sorensen](#), [Jim Berg](#), [Changkyu Lee](#), [Tim Jarsky](#), [Jonathan Ting](#), [Susan M. Sunkin](#), [David Feng](#), [Costas A. Anastassiou](#), [Eliza Barkan](#), [Kris Bickley](#), [Nicole Blesie](#), [Thomas Braun](#), [Krissy Brouner](#), [Agata Budzillo](#), [Shiella Caldejon](#), [Tamara Casper](#), [Dan Castelli](#), [Peter Chong](#), [Kirsten Crichton](#), [Christine Cuhacian](#), [Tanya L. Daigle](#), [Rachel Dalley](#), [Nick Dee](#), ... [Christof Koch](#) [+ Show authors](#)

[Nature Neuroscience](#) **22**, 1182–1195 (2019) | [Cite this article](#)

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Abstract

Understanding the diversity of cell types in the brain has been an enduring challenge and requires detailed characterization of individual neurons in multiple dimensions. To systematically profile morpho-electric properties of mammalian neurons, we established a single-cell characterization pipeline using standardized patch-clamp recordings in brain slices and biocytin-based neuronal reconstructions. We built a publicly accessible online database, the Allen Cell Types Database, to display these datasets. Intrinsic physiological properties were measured from 1,938 neurons from the adult laboratory mouse visual cortex, morphological properties were measured from 461 reconstructed neurons, and 452 neurons had both measurements available. Quantitative features were used to classify neurons into distinct types using unsupervised methods. We established a taxonomy of morphologically and electrophysiologically defined cell types for this region of the cortex with 17